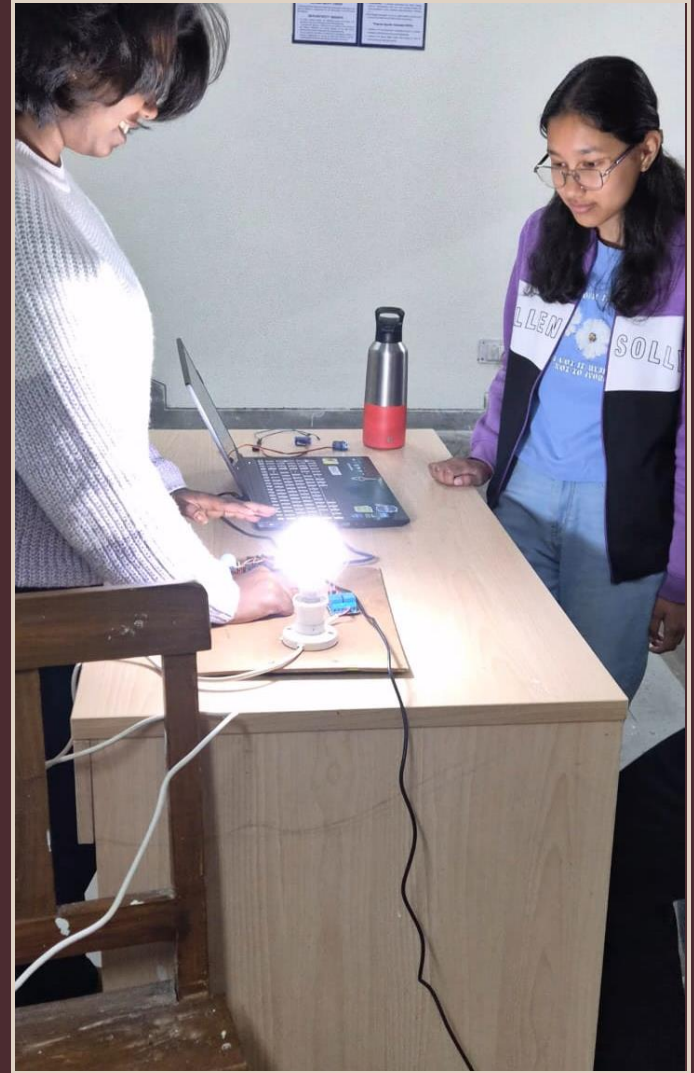


# Microcontrollers & Coding Basics

Conducted on 14th April 2026



# Introduction

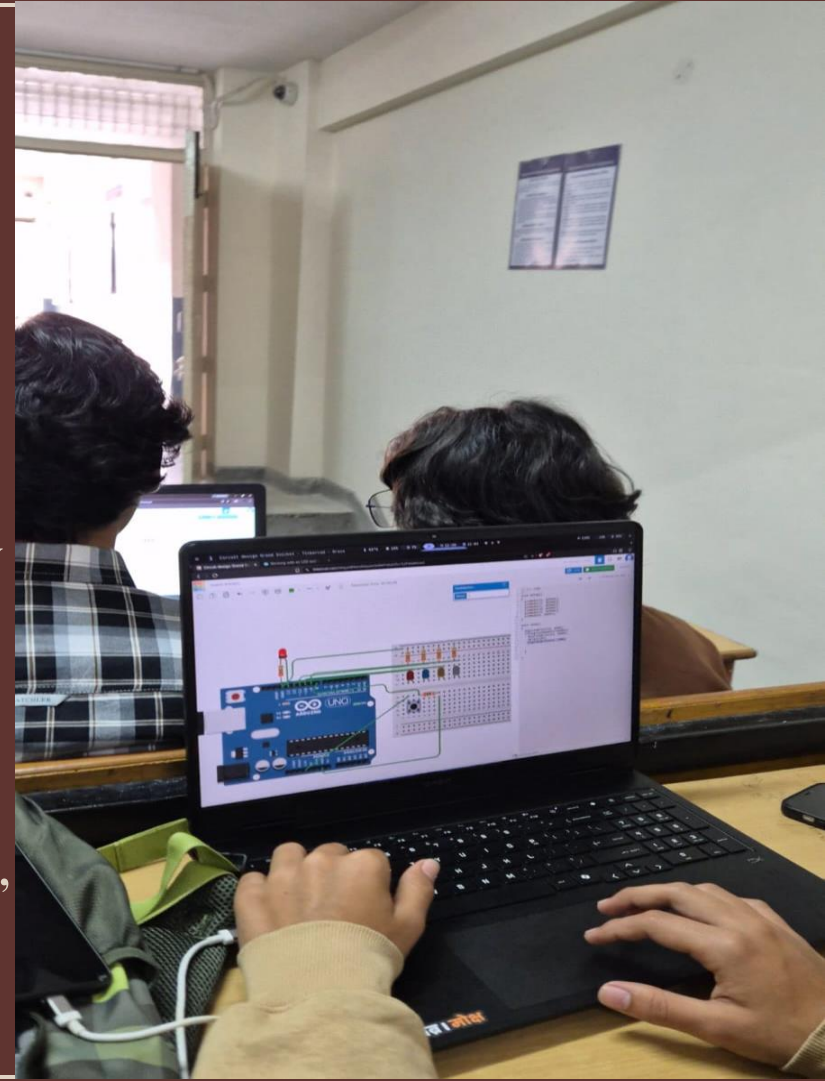
As detailed in the workshop document:

**Microcontroller Boards:** Handled through platforms like **Arduino UNO** and **Raspberry Pi**, focusing on hardware pinouts, internal architecture, and General Purpose Input/Output (GPIO) pin interfacing.

**Sensors & Input Modules:** Utilizes hardware such as ultrasonic distance sensors, optical sensors, dynamic display modules, and analog-to-digital conversion components.

**Embedded Programming:** Relies on core software control techniques including Pulse Width Modulation (PWM), sensor polling algorithms, and signal timing loops.

**Practical Applications:** Involves live breadboard prototyping, circuit design testing, direct code uploading, and dynamic hardware troubleshooting/debugging.





# Members

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14.Aman





## Key Topics:

Hardware pinouts and GPIO interfacing for Arduino UNO & Raspberry Pi; working with ultrasonic, optical, and display sensors; embedded C/C++ programming (PWM, polling, timing); and live breadboard prototyping



THANK YOU!